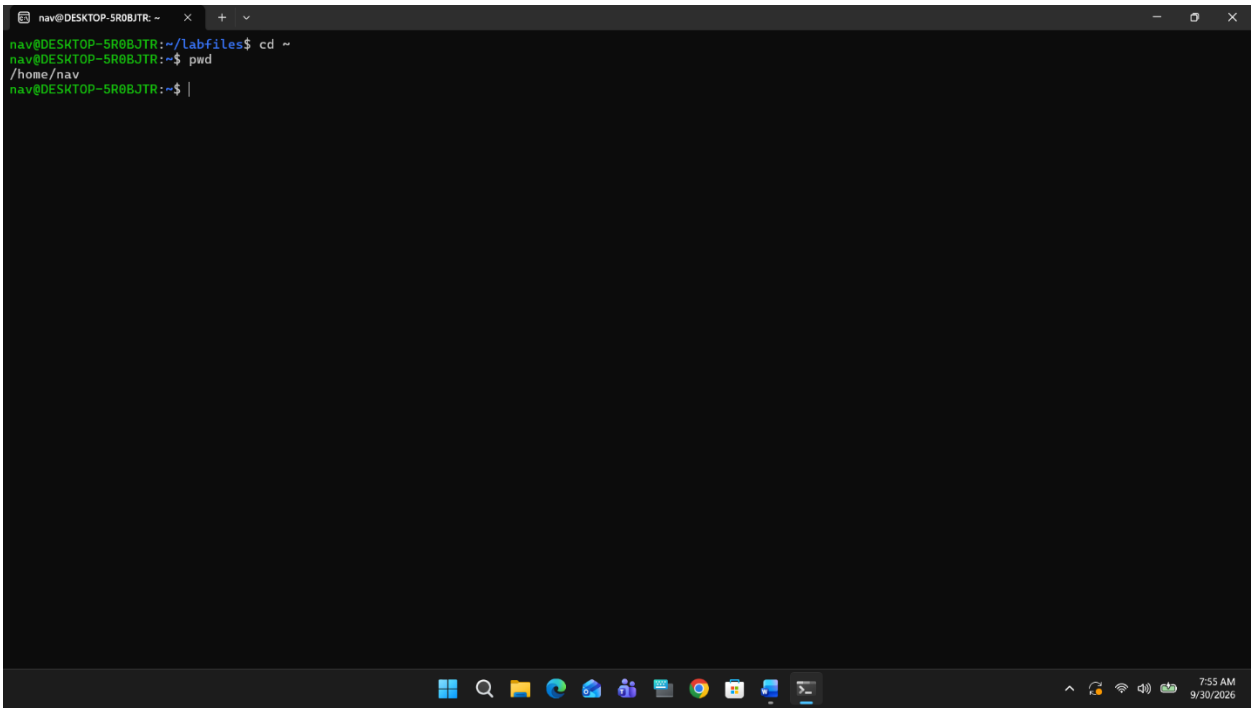
Linux System and Shell Scripting Lab – Lab 1

Question 3

Q.3. Create a College directory hierarchy with subdirectories (CSE, ECE, ME, Civil), nested semester/subject folders, and display it using tree, pwd, cd, mkdir, rmdir, ls.

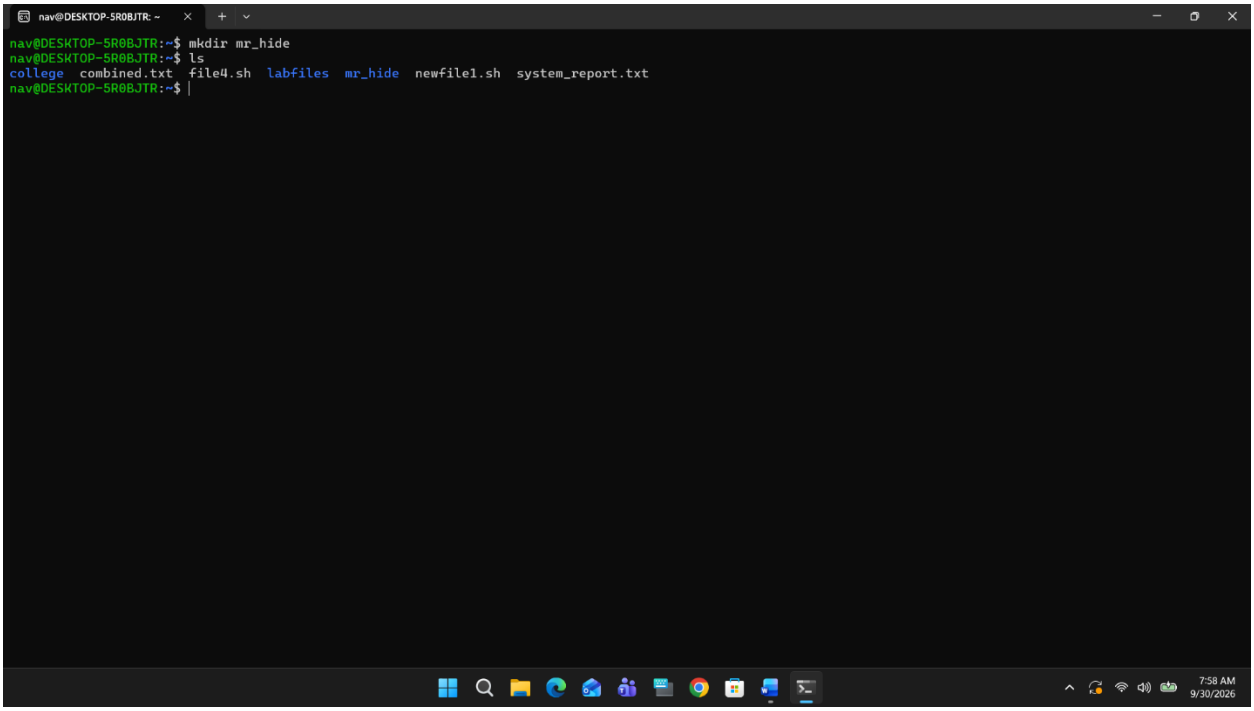
Step 1 – Return to the home directory and verify the location

We returned to the home directory using cd ~ and used pwd to verify the current location.



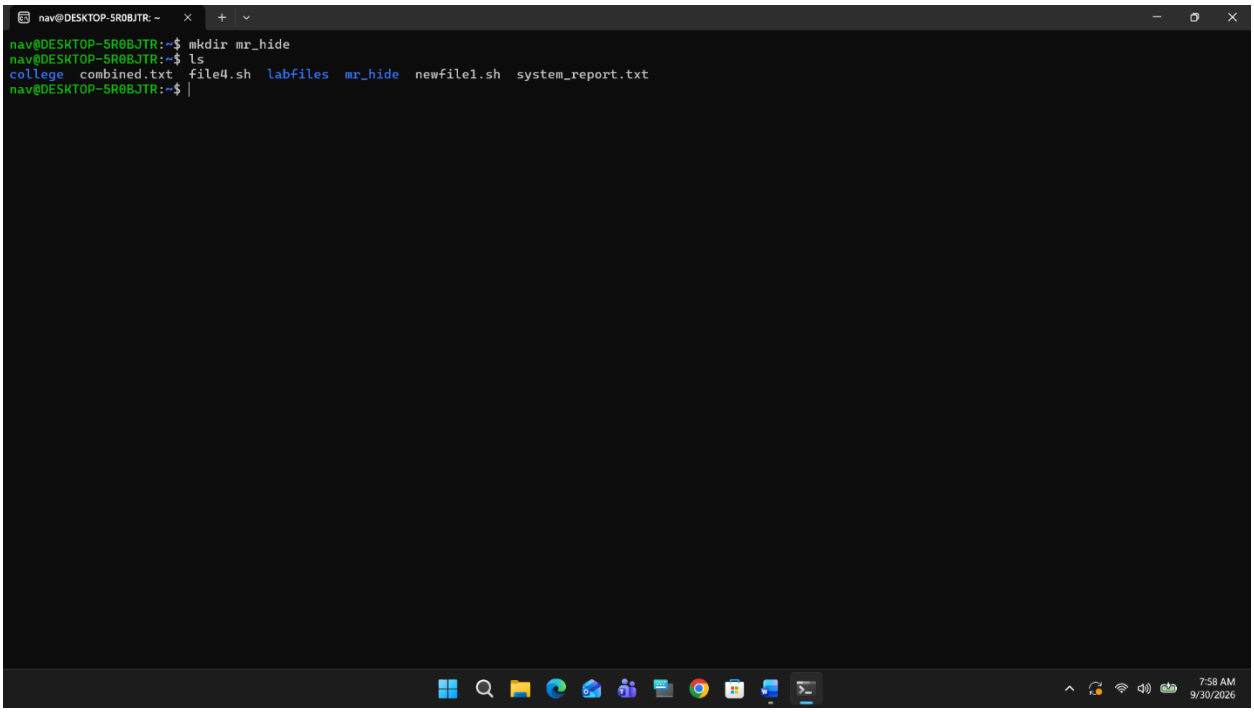
Step 2 – Create the College directory

We created the College directory inside Desktop to begin the required directory hierarchy.



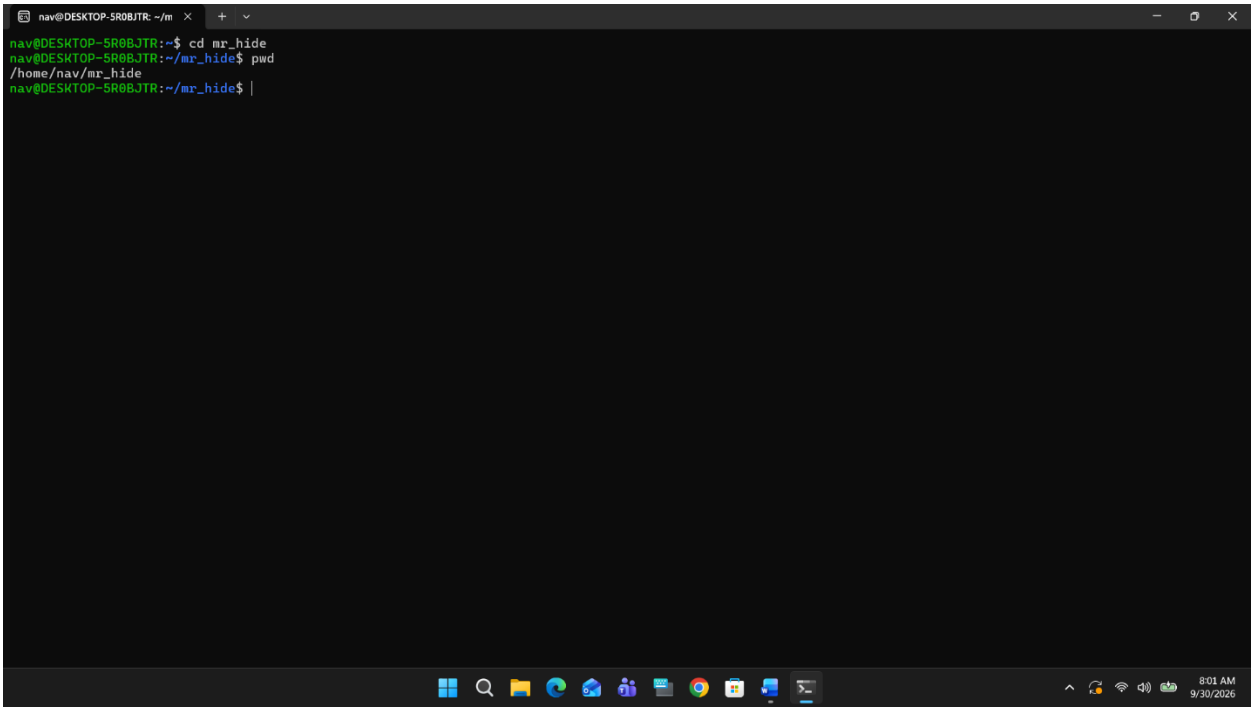
Step 3 – Verify the Desktop contents

We used ls to confirm that the College directory was created along with the existing LabFiles directory.



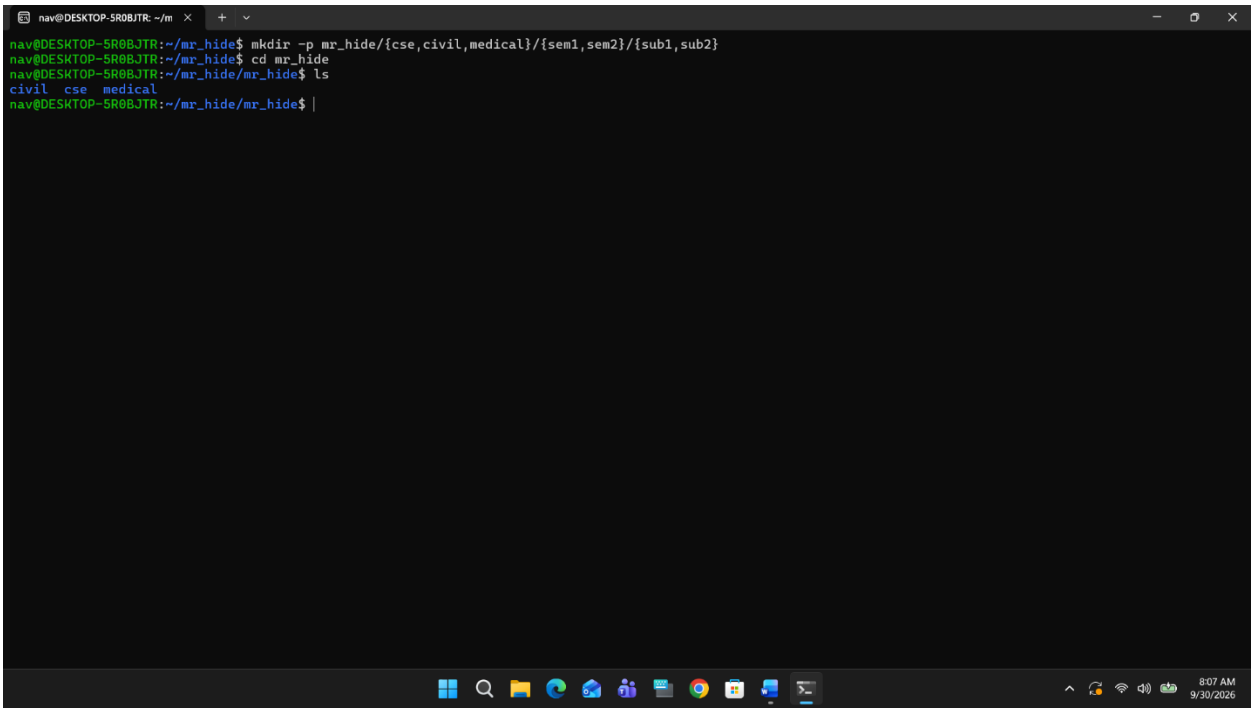
Step 4 – Enter the College directory

We used cd College/ to enter the College directory and continue creating its structure.



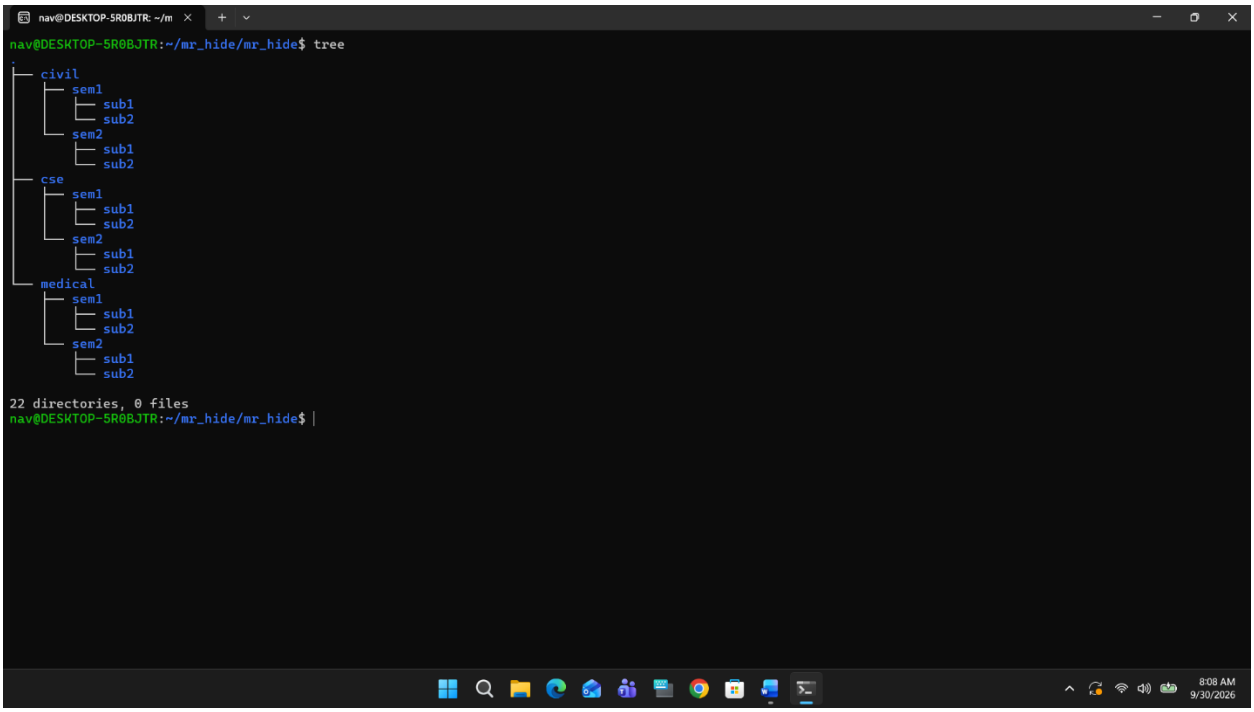
Step 5 – Create the department directories

We created the four department directories: CSE, ECE, ME and Civil. The directory listing confirmed that all four departments were created.



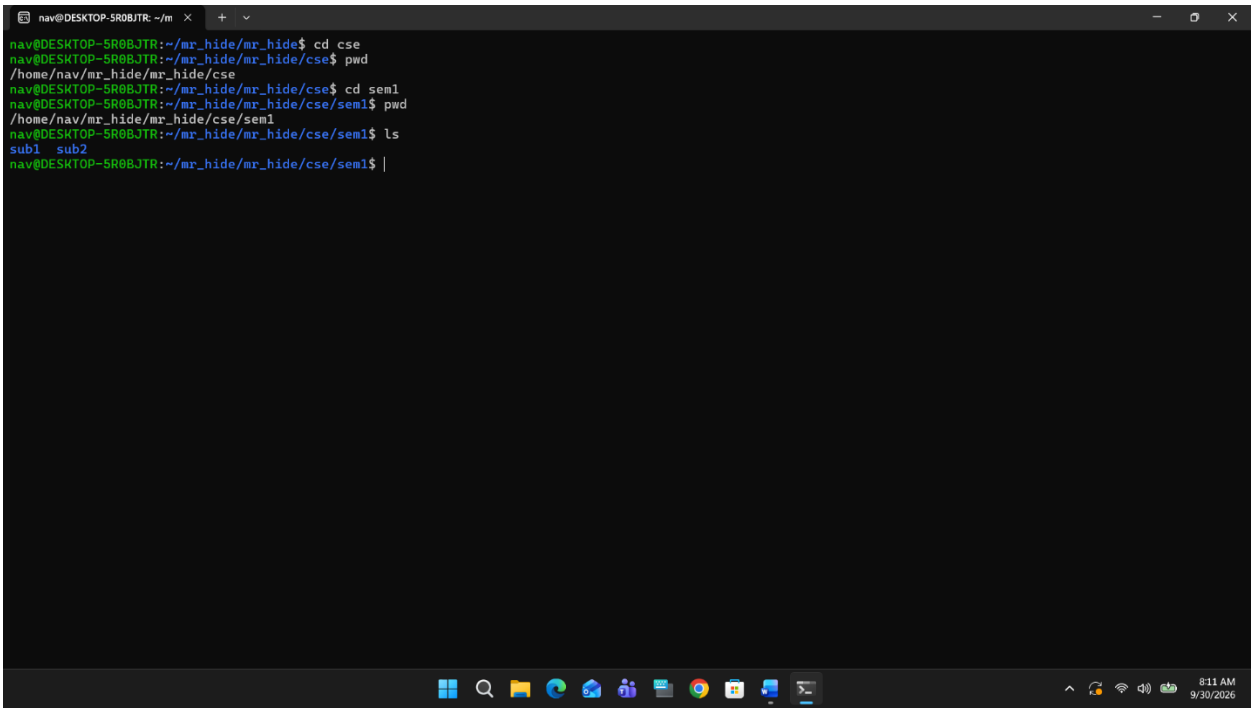
Step 6 – Display the complete department, semester and subject structure using tree

We used tree to display the complete College hierarchy, including all departments, semesters and subject folders.



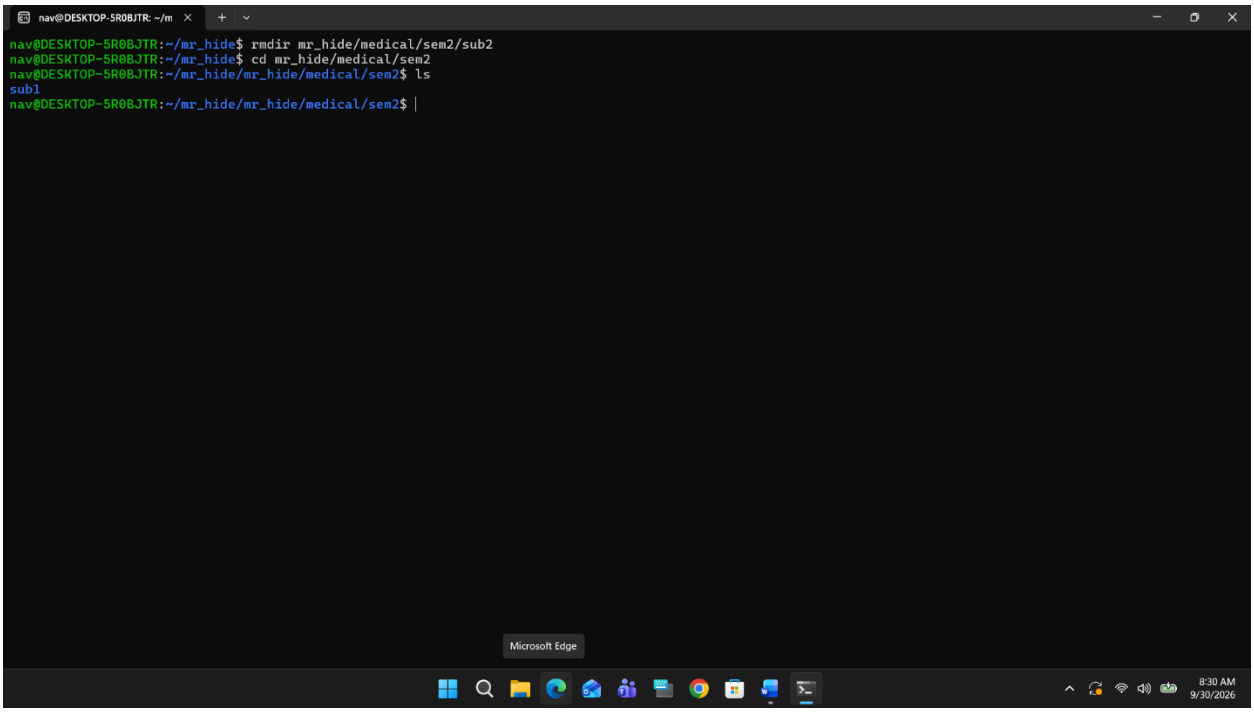
Step 7 – Enter CSE/Sem1 and verify the subjects

We entered the CSE/Sem1 directory using cd and used ls to verify that Subject1 and Subject2 were present.



Step 8 – Remove an empty subject directory using rmdir

We used rmdir to remove the empty College/Civil/Sem2/Subject2 directory and then used ls to confirm that only Subject1 remained in Sem2.



Step 9 – Display the recursive directory listing

We used ls -R College | head -20 to display the first 20 lines of the recursive College directory listing.

